PID Tuner (SistemaP2/Hipotenso/PID Controller) - Step Plot: Reference tracking

PID TUNER | VIEW

Plant: Plant | Inspect — PLANT

Type: PIDF | Form: Parallel | Options — CONTROLLER

Domain: Time | Add Plot — DESIGN

Slower — Response Time (seconds) — Faster: .0001391

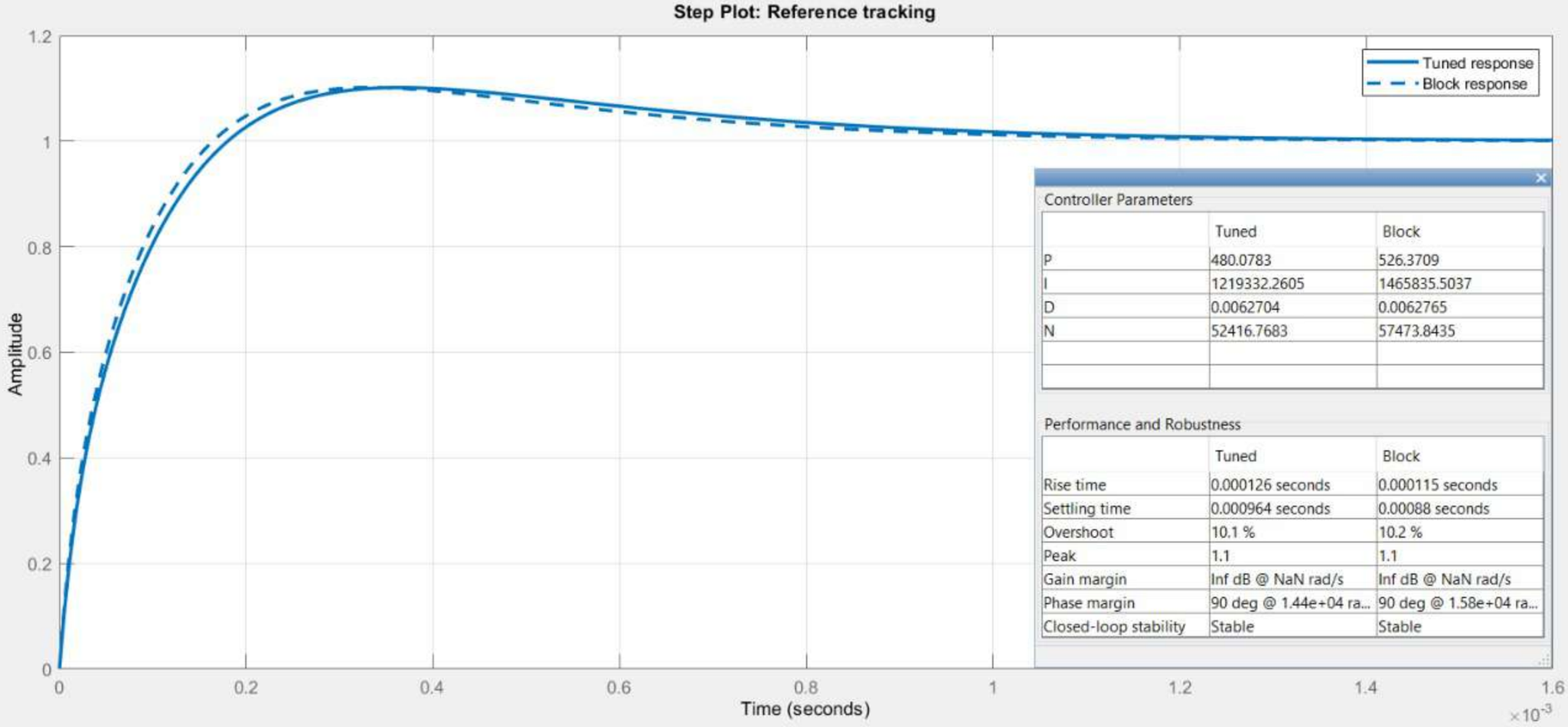
Aggressive — Transient Behavior — Robust: 0.9

Reset Design | Show Parameters — TUNING TOOLS

Update Block — RESULTS

Data Browser

Step Plot: Reference tracking



Controller Parameters

	Tuned	Block
P	480.0783	526.3709
I	1219332.2605	1465835.5037
D	0.0062704	0.0062765
N	52416.7683	57473.8435

Performance and Robustness

	Tuned	Block
Rise time	0.000126 seconds	0.000115 seconds
Settling time	0.000964 seconds	0.00088 seconds
Overshoot	10.1 %	10.2 %
Peak	1.1	1.1
Gain margin	Inf dB @ NaN rad/s	Inf dB @ NaN rad/s
Phase margin	90 deg @ 1.44e+04 ra...	90 deg @ 1.58e+04 ra...
Closed-loop stability	Stable	Stable

Controller Parameters: P = 480.1, I = 1.219e+06, D = 0.00627, N = 5.242e+04

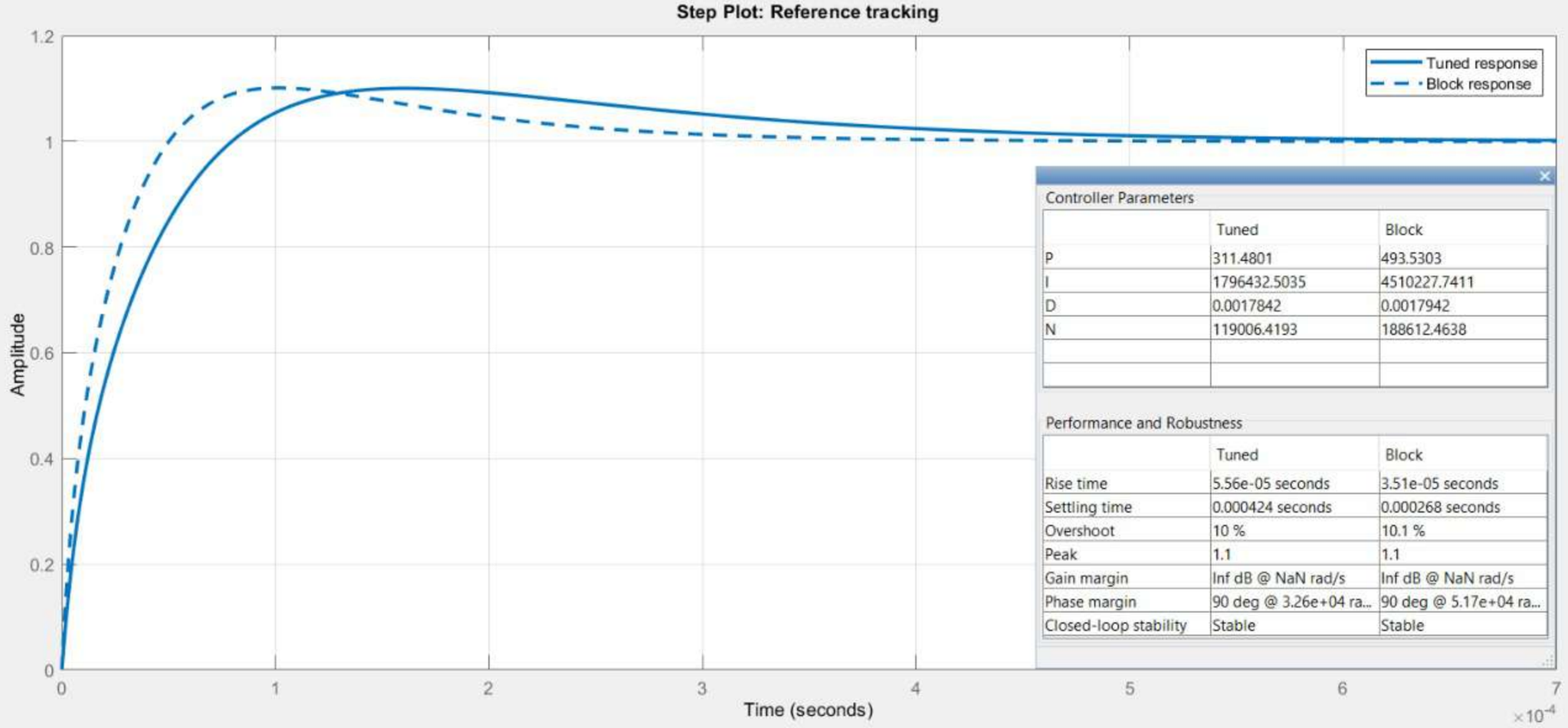
PID Tuner (SistemaP2/Hipertenso/PID Controller) - Step Plot: Reference tracking

PID TUNER | VIEW

Plant: Plant | Inspect — PLANT
Type: PIDF | Form: Parallel | Options — CONTROLLER
Domain: Time | Add Plot — DESIGN
Slower — Response Time (seconds) — Faster
Aggressive — Transient Behavior — Robust — TUNING TOOLS
0.0006129 | 0.9 | Reset Design | Show Parameters
Update Block — RESULTS

Step Plot: Reference tracking

Data Browser



Controller Parameters

	Tuned	Block
P	311.4801	493.5303
I	1796432.5035	4510227.7411
D	0.0017842	0.0017942
N	119006.4193	188612.4638

Performance and Robustness

	Tuned	Block
Rise time	5.56e-05 seconds	3.51e-05 seconds
Settling time	0.000424 seconds	0.000268 seconds
Overshoot	10 %	10.1 %
Peak	1.1	1.1
Gain margin	Inf dB @ NaN rad/s	Inf dB @ NaN rad/s
Phase margin	90 deg @ 3.26e+04 ra...	90 deg @ 5.17e+04 ra...
Closed-loop stability	Stable	Stable

Controller Parameters: P = 311.5, I = 1.796e+06, D = 0.001784, N = 1.19e+05